Wei You

Email: weiyou@ust.hk
Office: Room 5559D, HKUST
Last updated: May 21, 2025

Academic Appointment

Hong Kong University of Science and Technology

Assistant Professor (2019-present), Industrial Engineering and Decision Analytics (tenure track)

Education

Columbia University

Ph.D. in Operations Research (2019)

Thesis: A Robust Queueing Network Analyzer Based on Indices of Dispersion

Committee: Ward Whitt (Advisor), Henry Lam, Martin Reiman, David Yao, Assaf Zeevi

Nanjing University

B.Sc in Mathematics (2014)

Research Interests

Stochastic models, queueing theory, applied probability, sequential decision-making, and their applications to the performance analysis and design of service systems and digital experimentation.

Publications & Preprints

Preprints

1. Cost-Aware Structured Bandits,
   Xinyu Liu, Chao Qin and Wei You, working paper, 2025.
2. Transfer Learning with Partially Observable Offline Data via Causal Bounds,
   Xueping Gong, Wei You and Jiheng Zhang, Major revision at **Operations Research**, 2025.
3. Service Rate Differentiation for Homogeneous Impatient Customers,
   Chenguang (Allen) Wu and Wei You, Major revision at **Manufacturing & Service Operations Management**, 2024.
4. Robust Queueing for Queues with Impatient Customer,
   Wei You, working paper, 2024.
5. A Multi-Armed Bandit Approach to Online Resource Allocation for Service Platforms,
   Zihao Wang and Wei You, working paper, 2023.

Publications

1. Dual-Directed Algorithm Design for Efficient Pure Exploration
   Chao Qin and Wei You, to appear in **Operations Research**.

2. Stochastic Compositional Optimization with Compositional Constraints
   Shuoguang Yang, Wei You, Zhe Zhang, and Ethan X. Fang, to appear in **INFORMS Journal on Optimization**.

3. Information-Directed Selection for Top-Two Algorithms
   Wei You, Chao Qin, Zihao Wang, Shuoguang Yang, **Conference on Learning Theory (COLT)**, 2023.

4. New decomposition approximations for queueing network
   Ward Whitt and Wei You, **Queueing Systems**, 2022.

5. A Robust Queueing Network Analyzer Based on Indices of Dispersion
   Ward Whitt and Wei You, **Naval Research Logistics**, 2022.

6. Heavy-Traffic Limits for Stationary Network Flows
   Ward Whitt and Wei You, **Queueing Systems**, 2020.

7. Time-Varying Robust Queueing
   Ward Whitt and Wei You, **Operations Research**, 2019.

8. The Advantage of Indices of Dispersion in Queueing Approximations
   Ward Whitt and Wei You, **Operations Research Letters**, 2019.

9. Heavy-Traffic Limit of the GI/GI/1 Stationary Departure Process and its Variance Function
   Ward Whitt and Wei You, **Stochastic Systems**, 2018.

10. Using Robust Queueing to Expose the Impact of Dependence in Single-Server Queues
    Ward Whitt and Wei You, **Operations Research**, 2018.

Teaching Experience

Hong Kong University of Science and Technology, Instructor

IEDA5270 Engineering Statistics (2019-2025)

IEDA2540 Statistics for Engineers (2020-2025)

EEMT5500 Applied Probability, Statistics and Data Analytics (2022)

Columbia University, Teaching Assistant

Applied Statistical Models (2017), Simulation (2017, 2015), Stochastic Models (2014)

Conference & Invited Talks

Conferences

"Transfer Learning with Partially Observable Offline Data via Causal Bounds"
EcoSta 2025, Tokyo, Japan

"Cost-Aware Exploration for Structured Bandits"
2025 INFORMS Applied Probability Conference, Atlanta, GA
2025 INFORMS International Meeting, Singapore

"Dual-Directed Algorithm Design for Efficient Pure Exploration"
2024 INFORMS Annual Meeting, Seattle, WA
IASC-ARS Interim Conference 2024, Taipei City
JCSDS 2023, Kunming, Yunnan

"Information-Directed Selection for Pure Exploration"
2023 INFORMS Annual Meeting, Phoenix, AZ

"Information-Directed Selection for Top-Two Algorithms"
Conference on Learning Theory (COLT) 2023, Bangalore, India
2023 INFORMS APS Meeting, Nancy, France
POMS-HK 2023, Hong Kong

"Approximation for Single-Server Queues with Abandonment via Robust Queueing"
2024 INFORMS MSOM Conference, Minneapolis, Minnesota
2021 INFORMS Annual Meeting, Anaheim, CA

"Online Resource Allocation for Service Platforms"
Virtual 2020 INFORMS Annual Meeting

"Heavy-Traffic Limit for Stationary Network Flows"
2019 INFORMS APS Meeting, Brisbane, Australia

"Time-Varying Robust Queueing for Quantile Approxiamtion"
POMS-HK 2020, Hong Kong
2019 INFORMS APS Meeting, Brisbane, Australia

"A Robust Queueing Network Analyzer Based on Indices of Dispersion"
2018 INFORMS Annual Meeting, Phoenix, AZ

"A Robust Queueing Network Analyzer for a Series of Single-server Queues"
2017 INFORMS Annual Meeting, Houston, TX

"Time-Varying Robust Queueing"
2017 INFORMS APS Meeting, Evanston, IL

"Using Robust Queueing To Expose The Impact of Dependence In Single Server Queues"
2016 INFORMS Annual Meeting, Nashville, TN

Invited Talks

Antai College of Economics & Management, Shanghai Jiao Tong University, 2025

Institute of Statistics and Big Data, Renmin University of China, 2024

School of Statistics and Data Science, Nankai University, 2024

School of Management, Fudan University, 2024

School of Digital Economics, Shanghai University of Finance and Economics, 2023

School of Management and Engineering, Nanjing University, 2023

Institute of Applied Mathematics, Chinese Academy of Sciences, 2019

Institute of Mathematical Science, ShanghaiTech University, 2019

Industrial Engineering and Decision Analytics, HKUST, 2019

Industrial and System Engineering, Virginia Tech, 2019

System and Industrial Engineering, University of Arizona, 2019

Business School, University of Miami, 2019

School of Management, University of Science and Technology of China, 2019

Institute for Data and Decision Analytics, Chinese University of Hong Kong, Shen Zhen, 2019

System Engineering and Engineering Management, Chinese University of Hong Kong, 2019

School of Management and Engineering, Nanjing University, 2017

Grants

Hong Kong RGC-GRF (2024-2026). Project No. 16212823, PI, HK$651,464.

Hong Kong RGC-ECS (2021-2023). Project No. 26212320, PI, HK$529,972.

Students

Xinyu Liu, Ph.D. (2024-present)

Qiaoqiao Wang, Ph.D. (2023-present)

Ke Lu, Mphil. (2023-present)

Zihao Wang, Ph.D. (2019-2023)
— Thesis: "Online Resource Allocation for Service Platforms and Bandit Experiments"
— First placement: Equity Derivatives Quant, Bank of America.

Yingxian Wu, Mphil. (2020-2022), Co-advised with Prof. Sisi Jian
— Thesis: "Mix Allocation of Two Types of Passengers and Drivers"
— First placement: Investment Risk Management Assistant, E Fund Management Co., LTD.

Academic Service

Reviewer: Management Science, Operations Research, Queueing Systems, Stochastic Systems, Performance Evaluation, Production and Operations Management, Naval Research Logistics, Probability in the Engineering and Informational Sciences, Operations Research Letters, Winter Simulation Conference (2024, 2025), Algorithmic Learning Theory (2025), Israel Science Foundation (ISF)

Session Chair: 2017 INFORMS APS Meeting, POMS-HK 2023, 2025 INFORMS Applied Probability Conference, 2025 INFORMS International Meeting